

0.1 Checkpoint-verified neural configurations

The configurations below describe the circular-cylinder specialists analyzed for internal routing and parameter counts, and the centered-square specialists used to construct the routing/fusion candidates. They are not a uniform architecture shared across benchmarks. Unresolved baseline configurations are not inferred from these models.

Table 1: Verified specialist architectures. D is the input-feature dimension, h the encoder width, and m/B the expert expansion width/block count. Each group has six routed experts and one shared expert *per channel*. Both channels use Top-2 routed experts.

Specialist	(r_u, r_p)	D	h	m/B	Groups	Group selection
Circular Hopf	(32, 32)	493	256	1024/4	1	Frozen single group
Circular Periodic	(32, 32)	501	224	768/3	3	Learned Top-1
Square Steady	(5, 4)	91	224	768/3	3	Learned Top-1
Square Hopf	(11, 11)	178	256	1024/4	1	Frozen single group
Square Periodic	(28, 26)	431	224	768/3	3	Fixed group ID 1

Table 2: Source-level block definitions for the verified structured specialists. $s = r_u$ for velocity and $s = r_u + r_p$ for pressure; $o = r_u$ or r_p . Layer normalization is denoted LN. Ordinary Linear layers include bias unless explicitly stated otherwise.

Block	Dimensions	Operations
Encoder	$D \rightarrow h \rightarrow h \rightarrow h$	LN and SiLU after each Linear; dropout .04 after the first two activations.
Refinement	Two blocks, $h \rightarrow h \rightarrow h$	Pre-LN, SiLU, dropout .04; additive residual connection.
Expert nonlinear	$(h + s) \rightarrow h$; B blocks $h \rightarrow m \rightarrow h$; $h \rightarrow o$	LN/GELU input projection; pre-LN residual FFNs with GELU, dropout .04 and learned scalar residual scale (initially .1); LN output head.
Expert linear	Weight $o \times s$	Applied to the standardized local state slice; no bias.
Expert bilinear	$U, V \in \mathbb{R}^{o \times 4 \times s}$	Output-specific sum of four products, scaled by .05; no separate output-projection tensor or bias.
Routers	$(D + h) \rightarrow h \rightarrow G$ or 6	Pre-LN, SiLU, dropout .04; temperature-softmax and normalized Top- k .
Pressure gate	$h \rightarrow 64 \rightarrow r_p$	GELU and output sigmoid in adaptive-gate mode; componentwise. Square Steady instead stores a frozen, bypassed $224 \rightarrow 56 \rightarrow 2$ head.

Chart-specific execution. Square Steady freezes its velocity experts. The square Hopf candidate uses direct learned velocity derivatives and direct pressure outputs in its final rollout wrapper, although projected quantities remain input features. Its stored pressure gate is not used to assemble the pressure output. In the adaptive-gate paths, pressure is a gated Poisson base plus a learned residual, not a convex mixture of two independently predicted candidates.

Table 3: Verified centered-square outer networks. E2 and the two overlap gates have separately trained weights. The full and μ -only gates have identical architecture; μ -only masks eight standardized history channels.

Component	Dimensions	Activation	Trainable
E2	$1 \rightarrow 24 \rightarrow 3$	Tanh; output softmax	123
T2-C full	$9 \rightarrow 64 \rightarrow 64 \rightarrow 1$	SiLU; logit correction then sigmoid	4,865
T2-C μ -only	$9 \rightarrow 64 \rightarrow 64 \rightarrow 1$	Same network, masked history	4,865

E2 consumes only the standardized Reynolds number. T2-C consumes that parameter and eight standardized physical-history descriptors. Its final Linear layer is initialized to zero, so the initial gate reproduces the selected E2 pair preference. One scalar fusion weight is fixed for the entire window and shared by velocity and pressure; no fused field is fed back to either specialist.

Table 4: Specialist optimization settings from checkpoint configurations and their training entry points. All use AdamW and norm clipping at 1. “Budget” is the configured limit, not a claim that the selected checkpoint completed every curriculum stage.

Specialist	Initial LR / decay	Batch	Budget / selected	Curriculum / seed
Circular Hopf	.001/ 10^{-4}	232	8000/7200 steps	1, 2, 4, 8; 1248
Circular Periodic	.00055/.00015	256; rollout 2	240/85 epochs	4, 8, 12, 16; 1600
Square Steady	$1.5516 \times 10^{-5}/10^{-4}$	$16 \times 4 = 64$	4000/3200 steps	1, 4, 8, 16; 202607281
Square Hopf	.001/ 10^{-4}	46	8000/7200 steps	1, 2, 4, 8; 1248
Square Periodic	.00055/.00015	1024; rollout 56/56/40/28	720/490 epochs	4, 8, 12, 16; 1600

All five use cosine learning-rate schedules. Square Steady has a second parameter group at 0.1 times the listed initial learning rate. Hopf training uses fluctuation/normal-form losses and validation safeguards in addition to the common losses. Periodic checkpoints use the validation score combining RHS, velocity-head, and pressure-head relative errors, with weights 1, 1, 0.35. These objectives and selectors are not uniform across charts.

Table 5: Centered-square outer-network training. All use AdamW with initial learning rate 3×10^{-4} , weight decay 10^{-4} , gradient-norm clipping at 1, and validation every 100 steps (also at the first/final step).

Component	Batch / steps	Schedule / seed	Selection
E2	110 /8000	Cosine to 1.5×10^{-5} ; 20260730	Maximize validation balanced accuracy plus macro F1 minus .001 times the current training-batch loss.
T2-C SH	56 /8000	Constant; 42001/2/3	Lexicographically minimize validation worst-Re mean and then global mean joint error.
T2-C HP	32 /8000	Constant; 42001/2/3	Same selector and matched full/ μ -only settings.

E2 minimizes class-weighted cross entropy plus an adjacency penalty $0.05 \text{ mean}(\pi_S \pi_P)$.

T2-C minimizes the mean summed squared relative velocity/pressure field errors across the training windows and all 24 prediction steps. Its validation selector uses the corresponding unsquared component errors. The three gate seeds are 42001, 42002, and 42003.

Table 6: Recomputed trainable counts for the verified source models. Active counts are independently traced only for the circular Hopf/Periodic prediction paths. A dash denotes not independently verified, not zero. Fixed tensors and frozen parameters are excluded from trainable counts.

Specialist	Trainable	Frozen parameters	Active prediction
Circular Hopf	31,810,392	193,755	14,165,816
Circular Periodic	48,617,547	0	8,308,959
Square Steady	23,746,370	22,908,825	—
Square Hopf	31,205,844	112,485	—
Square Periodic	48,209,967	0	—

Active counts refer to unique trainable parameters consumed by one batch-size-one prediction-path routing decision, not a union over a batch, trajectory, or four RK stages. They are not FLOP counts, timings, or speedups. Unresolved circular Steady, Vanilla-FNN-MoE, DataOnly-MoE, Global MoE, and Pinball publication identities are intentionally excluded from these tables.